Roy Howard Goodman
goodman@njit.edu
https://roygoodman.net

Professor
Department of Mathematical Sciences
New Jersey Institute of Technology
Newark, NJ 07102
(973) 642-4261

Roy Goodman

Education

1994–1999 • Courant Institute of Mathematical Sciences, New York University, New York, NY
Ph.D. Mathematical Sciences (David W. McLaughlin, dissertation advisor)

1990–1994 • University of Michigan, Ann Arbor, MI
B.S. Mathematics (honors option), with highest honors

Research Interests

- Linear and nonlinear wave propagation, PDE
- Dynamical systems, invariant manifold computations
- Mathematical modeling and asymptotic analysis of physical systems
- Vortex dynamics
- Nonlinear phenomena in optics
- Numerical simulation

Professional Experience

2024– • **Professor**, New Jersey Institute of Technology
2022– • **Associate Chair for Graduate Studies**, NJIT Department of Mathematical Sciences
2008–2024 • **Associate Professor (Tenured)**, New Jersey Institute of Technology
2002–2008 • **Assistant Professor**, New Jersey Institute of Technology
2001–2002 • **Visiting Assistant Professor**, New Jersey Institute of Technology
1999–2001 • **Visiting Research Fellow**, Princeton University and Bell Laboratories (Lucent), under NSF University-Industry Cooperative Research Program in the Mathematical Sciences (advisors Philip Holmes, Princeton, and Michael Weinstein, Bell Labs)

Visiting Positions

Spring 2026 • **Long-Term Visitor**, Department of Mathematics, Faculty of Nuclear Sciences and Physical Engineering, Czech Technical University in Prague
2018–2019 • **Long-Term Visitor**, Department of Mechanical Engineering, NYU Tandon School of Engineering
Fall 2016 • **Long-Term Visitor**, Institute for Mathematics and its Applications, University of Minnesota
2010–2011 • **Visiting Associate Professor**, Department of Mathematics, Technion – Israel Institute of Technology

Scholarly Activities

Refereed Journal Papers

2026 • T. Ohsawa, A. Richaud, and R. H. Goodman, *Small-Mass Asymptotics of Massive Point Vortex Dynamics in Bose–Einstein Condensates I: Averaging and Normal Forms*, submitted.

2026 • A. Anurag and R. H. Goodman, *Phase portraits and the bifurcation set for the three-vortex interaction system*, Commun. Nonlinear Sci. Numer. Simul. **163**, 110428.

2025 • R. H. Goodman, G. Conte, and J. L. Marzuola, *QGLAB: A MATLAB Package for Computations on Quantum Graphs*, SIAM J. Sci. Comput. **47**, B428–B453.

Scholarly Activities (continued)

2024 • A. Anurag, R. H. Goodman, and E. K. O'Grady, *A new canonical reduction of three-vortex motion and its application to vortex-dipole scattering*, Phys. Fluids **36**, 067110.

2023 • J. Adriazola, R. H. Goodman, and P. G. Kevrekidis, *Efficient Manipulation of Bose–Einstein Condensates in a Double-Well Potential*, Commun. Nonlinear Sci. Numer. Simul. **122**, 107219.

2023 • J. Adriazola and R. H. Goodman, *Apodizer Design to Efficiently Couple Light into a Fiber Bragg Grating*, SIAM J. Appl. Math. **83**, 1126–1145.

2023 • R. H. Goodman and B. M. Behring, *Transition to instability of the leapfrogging vortex quartet*, Mech. Res. Commun. **128**, 104068.

2022 • J. Adriazola and R. H. Goodman, *An optimal control approach to gradient-index design for beam reshaping*, J. Opt. Soc. Am. A **39**, 907–915.

2022 • J. Adriazola and R. H. Goodman, *A reduction-based strategy for optimal control of Bose–Einstein condensates*, Phys. Rev. E **105**, 025311.

2020 • T. E. Faver, R. H. Goodman, and J. D. Wright, *Solitary waves in mass-in-mass lattices*, Z. Angew. Math. Phys. **71**, 197.

2020 • A. Sagiv, A. Ditkowski, R. H. Goodman, and G. Fibich, *Loss of physical reversibility in reversible systems*, Phys. D **410**, 132515.

2020 • R. H. Goodman and M. Porfiri, *Topological features determining the error in the inference of networks using transfer entropy*, Math. Eng. **2**, 34–54.

2019 • B. M. Behring and R. H. Goodman, *Stability of leapfrogging vortex pairs: A semi-analytic approach*, Phys. Rev. Fluids **4**, 124703.

2019 • A. Kairzhan, D. E. Pelinovsky, and R. H. Goodman, *Drift of spectrally stable shifted states on star graphs*, SIAM J. Appl. Dyn. Syst. **18**, 1723–1755.

2019 • R. H. Goodman, *NLS bifurcations on the bowtie combinatorial graph and the dumbbell metric graph*, Discrete Contin. Dyn. Syst. **39**, 2203–2232.

2017 • R. H. Goodman, *Bifurcations of relative periodic orbits in NLS/GP with a triple-well potential*, Phys. D **359**, 39–59.

2015 • R. H. Goodman, P. G. Kevrekidis, and R. Carretero-González, *Dynamics of vortex dipoles in anisotropic Bose–Einstein condensates*, SIAM J. Appl. Dyn. Syst. **14**, 699–729.

2015 • R. H. Goodman, A. Rahman, M. J. Bellanich, and C. N. Morrison, *A mechanical analog of the two-bounce resonance of solitary waves: Modeling and experiment*, Chaos **25**, 043109.

2015 • R. H. Goodman, J. L. Marzuola, and M. I. Weinstein, *Self-trapping and Josephson tunneling solutions to the nonlinear Schrödinger / Gross–Pitaevskii equation*, Discrete Contin. Dyn. Syst. **35**, 225–246.

2013 • J. K. Wróbel and R. H. Goodman, *High-order adaptive method for computing two-dimensional invariant manifolds of three-dimensional maps*, Commun. Nonlinear Sci. Numer. Simul. **18**, 1734–1745.

2011 • R. H. Goodman, *Hamiltonian Hopf bifurcations and dynamics of NLS/GP standing-wave modes*, J. Phys. A: Math. Theor. **44**, 425101.

2011 • R. H. Goodman and J. K. Wróbel, *High-order bisection method for computing invariant manifolds of two-dimensional maps*, Int. J. Bifurcation Chaos **21**, 2017–2042.

2010 • J. Bławzdziewicz, R. H. Goodman, N. Khurana, E. Wajnryb, and Y.-N. Young, *Nonlinear hydrodynamic phenomena in Stokes flow regime*, Phys. D **239**, 1214–1224.

2008 • Y.-N. Young, J. Bławzdziewicz, V. Cristini, and R. H. Goodman, *Hysteretic and chaotic dynamics of viscous drops in creeping flows with rotation*, J. Fluid Mech. **607**, 209–234.

2008 • R. H. Goodman, *Chaotic scattering in solitary wave interactions: A singular iterated-map description*, Chaos **18**, 023113.

2008 • R. H. Goodman and M. I. Weinstein, *Stability and instability of nonlinear defect states in the coupled mode equations—analytical and numerical study*, Phys. D **237**, 2731–2760.

2007 • R. H. Goodman and R. Haberman, *Chaotic scattering and the* n*-bounce resonance in solitary-wave interactions*, Phys. Rev. Lett. **98**, 104103.

2005 • R. H. Goodman and R. Haberman, *Kink-antikink collisions in the ϕ^4 equation: The* n*-bounce resonance and the separatrix map*, SIAM J. Appl. Dyn. Syst. **4**, 1195–1228.

2005 • R. H. Goodman and R. Haberman, *Vector-soliton collision dynamics in nonlinear optical fibers*, Phys. Rev. E **71**, 056605.

2004 • R. H. Goodman and R. Haberman, *Interaction of sine-Gordon kinks with defects: The two-bounce resonance*, Phys. D **195**, 303–323.

2004 • R. H. Goodman, P. J. Holmes, and M. I. Weinstein, *Strong NLS soliton-defect interactions*, Phys. D **192**, 215–248.

Scholarly Activities (continued)

2002 • R. H. Goodman, R. E. Slusher, and M. I. Weinstein, *Stopping light on a defect*, J. Opt. Soc. Am. B **19**, 1635–1652.

2002 • R. H. Goodman, P. J. Holmes, and M. I. Weinstein, *Interaction of sine-Gordon kinks with defects: Phase space transport in a two-mode model*, Phys. D **161**, 21–44.

2002 • R. H. Goodman, A. J. Majda, and D. W. McLaughlin, *Modulations in the leading edges of midlatitude storm tracks*, SIAM J. Appl. Math. **62**, 746–776.

2001 • R. H. Goodman, M. I. Weinstein, and P. J. Holmes, *Nonlinear propagation of light in one-dimensional periodic structures*, J. Nonlinear Sci. **11**, 123–168.

1995 • R. H. Goodman, D. S. Graff, L. M. Sander, P. Leroux-Hugon, and E. Clément, *Trigger waves in a model for catalysis*, Phys. Rev. E **52**, 5904–5909.

Book Chapters

2019 • R. H. Goodman, *Mathematical analysis of fractal kink-antikink collisions in the ϕ^4 model*, in **A dynamical perspective on the ϕ^4 model**, Springer, P. G. Kevrekidis and J. Cuevas-Maraver, eds.

Unrefereed Proceedings Publications

2005 • R. H. Goodman, R. E. Slusher, M. I. Weinstein, and M. Klaus, *Trapping light with grating defects*, in **Mathematical Methods for Nonlinear Wave Propagation**, Contemp. Math. **379**, 83–92.

2005 • P. J. Holmes, R. H. Goodman, and M. I. Weinstein, *Trapping of kinks and solitons by defects: Phase space transport in finite-dimensional models*, Proceedings of the International Conference on Progress in Nonlinear Science dedicated to Alexander Andronov, Nizhny Novgorod, Russia, July 2001.

Book Reviews

2018 • Review of *Methods of Mathematical Modeling* by T. Witelski and M. Bowen, SIAM Rev. **60**, 215–216.

Other Unrefereed Writing

2024 • R. H. Goodman, *An English Translation of Gröbli's Ph.D. Dissertation: "Specielle Probleme über die Bewegung geradliniger paralleler Wirbelfäden"*, arXiv preprint, April 2024.

2024 • *SIAM New York–New Jersey–Pennsylvania Section Holds Inaugural Conference*, SIAM News, January 2024.

2019 • *Four Decades of Kink Interactions in Nonlinear Klein-Gordon Models: A Crucial Typo, Recent Developments and the Challenges Ahead*, DSWeb, a website of the SIAM Activity Group on Dynamical Systems, October 2019.

2019 • *Markdown: A Writing Tool for Every Applied Mathematician's Toolbox*, SIAM News, May 2019.

Teaching Publications

2007 • B. Bukiet and R. H. Goodman, Methods of Applied Mathematics (sample honors syllabus), Honors in Practice **3**, 171–175.

Conference Presentations, invited

6/2025 • Dynamics of Coherent Structures in Discrete and Continuum Nonlinear Systems, BIRS-IMAG, Granada, Spain

5/2025 • SIAM Conference on Dynamical Systems, Denver, CO, minisymposium speaker

4/2025 • Waves 2025, Athens, GA, minisymposium speaker

4/2023 • Second Drexel University Waves Workshop, Philadelphia, PA, invited speaker

8/2022 • SIAM Conference on Nonlinear Waves and Coherent Structures, Bremen, Germany, minisymposium speaker

7/2022 • Coherent Structures: Current Developments and Future Challenges, Lorentz Center, Leiden, Netherlands

3/2022 • Twelfth IMACS International Conference on Nonlinear Evolution Equations and Wave Phenomena, Athens, GA, minisymposium speaker

12/2019 • Canadian Mathematical Society Winter Meeting, Toronto, ON, invited speaker in session on Symmetry in Dynamical Systems

4/2019 • The Eleventh IMACS International Conference on Nonlinear Evolution Equations and Wave Phenomena: Computation and Theory, Athens, GA, minisymposium speaker

10/2018 • AMS Sectional Conference, Ann Arbor, MI, special session speaker

Scholarly Activities (continued)

6/2018 • SIAM Conference on Nonlinear Waves and Coherent Structures, Anaheim, CA, minisymposium speaker
8/2017 • Applied Mathematics, Modeling and Computational Science (AMMCS), Waterloo, ON, minisymposium speaker
5/2017 • SIAM Conference on Applications of Dynamical Systems, Snowbird, UT, minisymposium speaker
10/2016 • Workshop on Mathematical and Physical Models of Nonlinear Optics, Institute for Mathematics and Applications, Minneapolis, MN, invited speaker
6/2016 • Coherent Structures in PDEs and Their Application, Oaxaca, MX, invited speaker
7/2015 • Workshop on Pattern Formation, Dalhousie University, Halifax, NS, invited speaker
6/2015 • International Conference on Mathematics of Nonlinearity in Neural and Physical Science, Shanghai, China, invited speaker
8/2014 • SIAM Conference on Nonlinear Waves and Coherent Structures, Cambridge, UK, minisymposium speaker
5/2013 • Frontiers in Applied Mathematics, Newark, NJ, minisymposium speaker
5/2013 • SIAM Conference on Applications of Dynamical Systems, Snowbird, UT, featured minisymposium speaker
7/2012 • 2nd Conference on Localized Excitations in Nonlinear Complex Systems (LENCOS'12), Seville, Spain
4/2012 • Nonlinear Waves: Asymptotic Theory and Applied Mathematics, Mexico City, MX, invited presentation
11/2011 • SIAM Conference on Analysis of Partial Differential Equations, San Diego, CA, invited minisymposium speaker
8/2010 • SIAM Conference on Nonlinear Waves and Coherent Structures, Philadelphia, PA, minisymposium speaker
8/2009 • Analysis of nonlinear wave equations and applications in engineering, Banff International Research Station, Alberta, Canada, invited participant and speaker in 5-day workshop
5/2008 • Seventh AIMS Conference on Dynamical Systems and Differential Equations, Arlington, TX, minisymposium speaker
3/2008 • AMS Sectional Meeting, New York, NY, minisymposium speaker
5/2007 • SIAM Conference on Applications of Dynamical Systems, Snowbird, UT, minisymposium speaker
4/2007 • AMS Sectional Meeting, Hoboken, NJ, minisymposium speaker
12/2006 • CMS Winter Meeting, Toronto, ON, minisymposium speaker
9/2006 • SIAM Conference on Nonlinear Waves and Coherent Structures, Seattle, WA, minisymposium speaker
7/2006 • SIAM Annual Meeting, Boston, MA, minisymposium speaker
5/2005 • SIAM Conference on Applications of Dynamical Systems, Snowbird, UT, minisymposium speaker
4/2005 • IMACS International Conference on Nonlinear Evolution Equations and Wave Phenomena, Athens, GA, minisymposium speaker
10/2004 • SIAM Conference on Nonlinear Waves and Coherent Structures, Orlando, FL, minisymposium speaker
7/2004 • Workshop on Mathematical Ideas in Nonlinear Optics, Edinburgh, UK, invited 30-minute talk
10/2003 • AMS Sectional Meeting, Chapel Hill, NC, minisymposium speaker
7/2002 • SIAM 50th anniversary Conference, Philadelphia, PA, minisymposium speaker

Conference Presentations, Contributed

10/2024 • SIAM New York/New Jersey/Pennsylvania Section Conference, Rochester, NY, contributed talk
5/2023 • SIAM Conference on Applications of Dynamical Systems, Portland, OR, contributed talk
6/2012 • SIAM Conference on Nonlinear Waves and Coherent Structures, Seattle, WA, contributed talk
7/2011 • 10th International Conference on the Mathematical and Numerical Aspects of Waves, Vancouver, BC, contributed talk (refereed)
5/2010 • Frontiers in Applied Mathematics, Newark, NJ, minisymposium speaker and poster
5/2009 • SIAM Conference on Applications of Dynamical Systems, Snowbird, UT, poster with graduate student J. Wróbel
2/2009 • SIAM Conference on Computational Science and Engineering, Miami, FL, poster with graduate student J. Wróbel
7/2008 • SIAM Conference on Nonlinear Waves and Coherent Structures, Rome, Italy, contributed talk
5/2007 • Frontiers in Applied and Computational Mathematics, Newark, NJ, poster
5/2006 • Frontiers in Applied and Computational Mathematics, Newark, NJ, poster
10/2005 • International Workshop on Applied Dynamical Systems, Centre de recherches mathématiques, Montréal, QC, Canada, poster
5/2005 • Frontiers in Applied and Computational Mathematics, Newark, NJ, contributed poster
10/2004 • Conference in honor of D. McLaughlin's 60th birthday, Chapel Hill, NC, poster
5/2004 • Frontiers in Applied and Computational Mathematics, Newark, NJ, poster
1/2004 • Dynamics Days, Chapel Hill, NC, poster

Scholarly Activities (continued)

5/2003 • SIAM Conference on Applications of Dynamical Systems, Snowbird, UT, contributed talk
5/2002 • NSF-CBMS Regional Research Conference on Mathematical Methods for Nonlinear Wave Propagation, North Carolina A&T State University, poster

Summer Schools, Etc.

6/2024 • Summer School on Nonlinear Quantum Graphs, five lectures, Université Polytechnique Hauts-de-France, Valenciennes, France

Other Significant Talks

3/2024 • Mathematics Colloquium, Lehigh University
10/2020 • Mathematical Physics Seminar, Yeshiva University
2/2020 • Applied Mathematics Colloquium, New Jersey Institute of Technology
3/2019 • Applied Math Seminar, Drexel University
10/2018 • Analysis and PDE Seminar, University of North Carolina
4/2018 • Mathematics Seminar, University of Vermont
2/2017 • AIMS Seminar, University of Michigan
9/2016 • IMA Visitors Seminar, University of Minnesota
3/2015 • Math Department Seminar, Southern Methodist University
5/2014 • Dynamical Systems Seminar, Mechanical Engineering, NYU Polytechnic Institute
4/2014 • Mathematics Seminar, Montclair State University
4/2014 • Computational and Applied Mathematics Seminar, Rutgers University
3/2013 • Center for Computational Science Seminar, Tulane University
1/2013 • Applied Math Seminar, Drexel University
3/2012 • Center for Applied Mathematics Seminar, University of Massachusetts
12/2010 • Solid State Center Colloquium (Physics), Technion – Israel Institute of Technology
11/2010 • Applied Math Seminar, Weizmann Institute, Rehovot, Israel
10/2010 • Applied Math Colloquium, Tel Aviv University
10/2010 • Applied Math and PDE Seminar, Technion – Israel Institute of Technology
3/2009 • Mathematics Colloquium, University at Buffalo
1/2009 • Dynamical Systems Seminar, Drexel University
12/2007 • Lefschetz Center for Dynamical Systems seminar, Brown University
11/2007 • Dynamical Systems and Nonlinear Science Colloquium, Georgia Tech
2/2007 • Dynamical Systems and Nonlinear Science Seminar, Princeton University
11/2006 • Applied Mathematics Colloquium, Columbia University
9/2004 • Mathematics Colloquium, University of Vermont
2/2004 • Dynamical Systems and Nonlinear Science Seminar, Princeton University
10/2002 • Mathematics Colloquium, Southern Methodist University
2/2002 • Mathematics Colloquium, Worcester Polytechnic Institute
1/2002 • Mathematics Colloquium, Drexel University
12/2001 • Mathematics Colloquium, University of Maryland Baltimore County
11/2001 • Lefschetz Center for Dynamical Systems seminar, Brown University
10/2001 • Applied Mathematics Colloquium, NJIT

Conferences & Minisymposia Organized

5/2025 • Minisymposium, SIAM Conference on Applications of Dynamical Systems, Denver, CO
10/2024 • Organizing Committee, Second annual conference of the SIAM New York/New Jersey/Pennsylvania Section, Rochester, NY
10/2023 • Lead Organizer, First annual conference of the SIAM New York/New Jersey/Pennsylvania Section, Newark, NJ
5/2021 • Minisymposium, SIAM Conference on Applications of Dynamical Systems, online
10/2017 • Waves, Spectral Theory & Applications—Part 2, Chapel Hill, NC
9/2015 • Organizing Committee, Conference on Waves, Spectral Theory & Applications, Princeton, NJ
5/2014 • Organizing Committee & 2 Minisymposia, Frontiers in Applied & Computational Mathematics, Newark, NJ
5/2013 • Minisymposium, Frontiers in Applied & Computational Mathematics, Newark, NJ
8/2010 • Organizing Committee, SIAM Conference on Nonlinear Waves & Coherent Structures, Philadelphia, PA
8/2010 • Minisymposium, SIAM Conference on Nonlinear Waves & Coherent Structures, Philadelphia, PA

Scholarly Activities (continued)

5/2010 • Special Session, AMS Spring Eastern Meeting, Newark, NJ
3/2008 • Special Session, AMS Spring Eastern Meeting, New York, NY
5/2007 • Minisymposium, SIAM Conference on Applications of Dynamical Systems, Snowbird, UT
5/2007 • Minisymposium, Frontiers in Applied & Computational Mathematics, Newark, NJ
5/2005 • Minisymposium, SIAM Conference on Applications of Dynamical Systems, Snowbird, UT
5/2005 • Minisymposium, Frontiers in Applied & Computational Mathematics, Newark, NJ
4/2005 • Minisymposium, IMACS International Conference on Nonlinear Evolution Equations & Wave Phenomena, Athens, GA
10/2004 • Organizer, Conference in honor of D. McLaughlin's 60th birthday, Chapel Hill, NC
5/2004 • Minisymposium, Frontiers in Applied & Computational Mathematics, Newark, NJ

Other Workshop Participation

6/2014 • The Thirtieth Annual Workshop on Mathematical Problems in Industry, NJIT, Newark, NJ

Grants

2022–2025 • Principal Investigator, Dynamics and scattering of vortices and vortex rings, NSF DMS-2206016, $300,000
2008–2013 • Principal Investigator, Nonlinear waves and dynamical systems, NSF DMS-0807284, $199,881
2007–2009 • Co-Principal Investigator, **CSUMS:** Research and Education in Computational Mathematics for Undergraduates in the Mathematical Sciences at NJIT, NSF DMS-0639270, $536,696
2005–2008 • Principal Investigator, Mathematical methods for wave interactions, NSF DMS-0506495, $85,000
2004–2007 • Investigator, Acquisition of computer cluster for the Center of Applied Mathematics and Statistics at NJIT, NSF DMS-040590, Major Research Instrumentation grant, $270,870
2002–2005 • Principal Investigator, Pulse propagation and capture in Bragg grating optical fibers, NSF DMS-0204881, $73,001

Patents Awarded

10/5/2004 • R. H. Goodman, M. I. Weinstein, and R. E. Slusher, *Trapping light pulses at controlled perturbations in periodic optical structures*, U.S. Patent No. 6,801,685

Teaching

At NJIT

Undergraduate • Calculus I, Honors Calculus 2, Calculus 3A, Differential Equations, Intermediate Differential Equations (Dynamical Systems), Linear Algebra, Honors Linear Algebra, Applied Numerical Methods, Advanced Applied Numerical Methods, Mathematical Methods for Scientists and Engineers, Mathematical Analysis I, Honors Methods of Applied Mathematics 1 & 2 (Capstone course), Mathematical Modeling, Complex Analysis, Partial Differential Equations
Master's • Numerical Methods for Computation
Ph.D. • Asymptotic Methods I, Advanced Ordinary Differential Equations, Numerical Methods I, Wave Propagation, Special Topics: Dynamical Systems

At NYU

Undergraduate • Precalculus Mathematics

Ph.D. Dissertation Advisor

2024– • Nan Zhou
Topic: Generalization of the Leapfrogging Orbit of Vortices

Teaching (continued)

2022–2025 • Atul Anurag
Topic: The Global Phase Space of the Three-Vortex Problem and Its Application to Vortex-Dipole Scattering

2019–2022 • Grace Conte, University of North Carolina (co-advisor; J. Marzuola, principal advisor)
Dissertation: *Numerical Analysis Of Linear And Nonlinear Schrödinger Equations On Quantum Graphs*

2017–2021 • Jimmie Adriazola
Dissertation: *Coherent Control of Dispersive Waves*

2016–2020 • Brandon Behring
Dissertation: *Dances and Escape of the Vortex Quartet*

2013–2016 • Casayndra Basarab
Dissertation: *Hamiltonian Bifurcations in Schrödinger Trimers*

2008–2011 • Jacek Wróbel
Dissertation: *High-order Adaptive Method for Computing Invariant Manifolds of Maps*

Other Student Supervision

2025 • Simon Eskin, Undergraduate Summer Research Project (Case Western Reserve University student)

2023–2024 • Ellison O'Grady, Provost Undergraduate Research and Innovation Summer Fellow

2021–2022 • Noah Roselli, Undergraduate Research Project

2010 • Kyle Mahady, Graduate Summer Research Project

2009–2010 • Casayndra Basarab and Priyanka Shah, CSUMS Undergraduate Research Project

2007–2008 • Matthew Peragine and Fatima Elgammal, CSUMS Undergraduate Research Project

2007 • Xiaoni Fang, Graduate Summer Research Project

2006 • Maciej Malej, Undergraduate Summer Research Project

2004– • Member of dissertation committees for D. Cargill, M. Chabane, Y. Chen, G. Conte (UNC–Chapel Hill), I. Jancigova, Y. Joshi, Y. Mileyko, A. Rahman, B. Ren, M. Storm

Service

University

2022– • Math Department Representative to Committee on Graduate Education

2022 • Member, Faculty Senate Executive Committee

2021–2022 • Faculty Senate Representative to Committee on Graduate Education

2019–2022 • Member, Faculty Senate

2015–2018 • Member, Honors College Bauder Scholarship Committee

2014–2017 • Member, University Senate Committee on Campus Life

2003–2005 • Member, NJIT Committee on Health and Safety

2002–2004 • Advisor to undecided CSLA freshmen

Department

2022– • Member, Curriculum Committee

2022– • Associate Chair for Graduate Studies

2021– • Chair, CAMS Membership Committee

2021–2022 • Chair, Computation in the Curriculum Committee

2020– • Chair, then member, Online Instruction Committee

2017–2018, 2021–2022 • Member, Hiring Committee

2008–2022 • Applied Math Undergraduate Advisor

2002–2003, 2011–2018 • Applied Mathematics Minor Advisor

2003–2010 • Undergraduate Math Club and Pi Mu Epsilon Honor Society Advisor

2007–2010 • Organizer, Wave Propagation Seminar

Editorial Board Membership

2023– • Frontiers in Photonics

Service (continued)

Peer Reviewing Activity

- **Panelist**, NSF Division of Mathematical Sciences
- **Tenure Reviewer**, one review
- **Grant Reviewer**, individual grants, NSF-DMS, Israel Science Foundation, MITACS (Canada)
- **Referee**, AIMS Mathematics, Analysis and Applications, Annales Henri Lebesgue, Applied Mathematics Letters, Chaos, Chaos, Solitons & Fractals, Communications in Nonlinear Science and Numerical Simulation, Discrete and Continuous Dynamical Systems B, Europhysics Letters, European Physical Journal Plus, Frontiers in Photonics, IMA Journal of Applied Mathematics, International Journal of Engineering Science, International Journal of Theoretical Physics, Journal of Computational and Applied Mathematics, Journal of Computational Physics, Journal of Engineering Mathematics, Journal of High Energy Physics, Journal of Lightwave Technology, Journal of Low Temperature Physics, Journal of Marine Science and Engineering, Journal of Modern Optics, Journal of Nonlinear Science, Journal of the Optical Society of America B, Journal of Physics A, Mathematics, Mathematics and Computers in Simulation, Modern Physics Letters B, Nonlinearity, Numerical Methods for Partial Differential Equations, Optics Express, Optics Letters, Physica D, Physical Review A, Physical Review D, Physical Review E, Physical Review Fluids, Physical Review Letters, Physics of Fluids, PLOS ONE, Proceedings of the American Mathematical Society, Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, Results in Applied Mathematics, Scientific Reports, SIAM Journal on Applied Dynamical Systems, SIAM Journal on Applied Mathematics, SIAM Journal on Mathematical Analysis, SIAM Textbook Publishing, SIAM Undergraduate Research Online (SIURO), Studies in Applied Mathematics, Water, Wave Motion

Professional Societies

2022–2024 • SIAM New York/New Jersey/Pennsylvania Section, Founding president
2017–2018 • SIAM Nonlinear Waves SIAG Martin Kruskal Lecturer Selection Committee
2009–2010 • Secretary, SIAM Activity Group on Nonlinear Waves and Coherent Structures
• American Mathematical Society (member)
• SIAM, Society for Industrial and Applied Mathematics (member)
• SIAM Activity Group on Dynamical Systems (member)
• SIAM Activity Group on Nonlinear Waves and Coherent Structures (member)

References

- **Prof. Alejandro Aceves**
  Department of Mathematics
  Southern Methodist University
  237 Clements Hall
  Dallas, TX 75275
  aaceves@smu.edu
- **Prof. M. Gregory Forest**
  Grant Dahlstrom Distinguished Professor
  Department of Mathematics
  CB 3250 Phillips Hall
  University of North Carolina at Chapel Hill
  Chapel Hill, NC 27599
  (919) 962-9606
  forest@amath.unc.edu

References (continued)

- **Prof. David W. McLaughlin**
  Silver Professor of Mathematics and Neural Science
  Courant Institute of Mathematical Sciences
  New York University
  1113 Warren Weaver Hall
  251 Mercer St.
  New York, NY 10012
  (212) 998-3077
  david.mclaughlin@nyu.edu
- **Prof. Michael I. Weinstein**
  Department of Applied Physics and Applied Mathematics
  Columbia University
  200 S.W. Mudd - MC4701
  New York, NY 10027
  (212) 854-3624
  miw2103@columbia.edu